

Fabrication of 3-terminal graphene devices for Oxford NanoSpin at ORNL

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General Info about the chip

Devices consist of 10 x 10 mm chips which are pre-patterned with contact pads and alignment marks. Each chip consists of array of devices (23x38), see figure 1a. Each device consists of a source, drain and gate electrode, see figure 1b. Gate and drain are interconnected in each column. The top of the chip can be identified by the group logo and horizontal line. In the GDS file, these features are shown in layer 1/0.

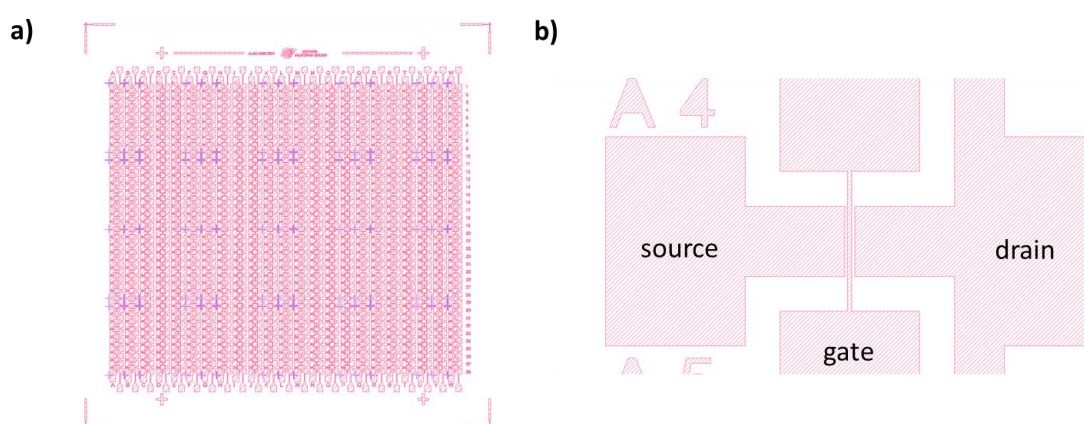


Figure 1 a) Diagram of the entire chip. b) Diagram of a device with connections labelled.

The following steps have been used to produce the features you see on the chip:

1. Fabrication of gate electrodes with UV lithography (5/25 nm Ti/Pd)
2. ALD of 10 nm HfO₂ as gate oxide.
3. Fabrication of source-drain contacts with UV lithography (5/60 nm Ti/Au)
4. HF etch to remove gate oxide on contact pads on gate electrode.
5. Transfer of monolayer CVD graphene across the entire wafer.
6. Spin coating of protective layer (AR-N-7500.18) and dicing into chips.
7. Removal of protective coating and debris.

All features have been patterned on the polished side of the wafer. The reverse of the substrate has no features. The substrate is a Si wafer with 300 nm of thermal oxide.

EBL fabrication notes

The aim of this work is to pattern the graphene layer into bowties/nano-constrictions using a negative resist mask and oxygen plasma etching. Please see "3T_graphene_ILG2.0_Alex_Gee.gds" for the pattern design. Each of the devices shown in figure 1b will have a nanoconstriction patterned on it. In the GDSII file, the nanoconstrictions are shown in layer 3/0, shown in figure 2.

Graphene is problematic as it easily becomes contaminated by resist residues which cannot be removed with plasma etching without damaging or removing the graphene layer. We find that contamination is minimized using AR-N 7500.18 and gives good results patterning the

nanoconstrictions. If possible, I would like to spin coat the chips and do the development and oxygen plasma etching myself.

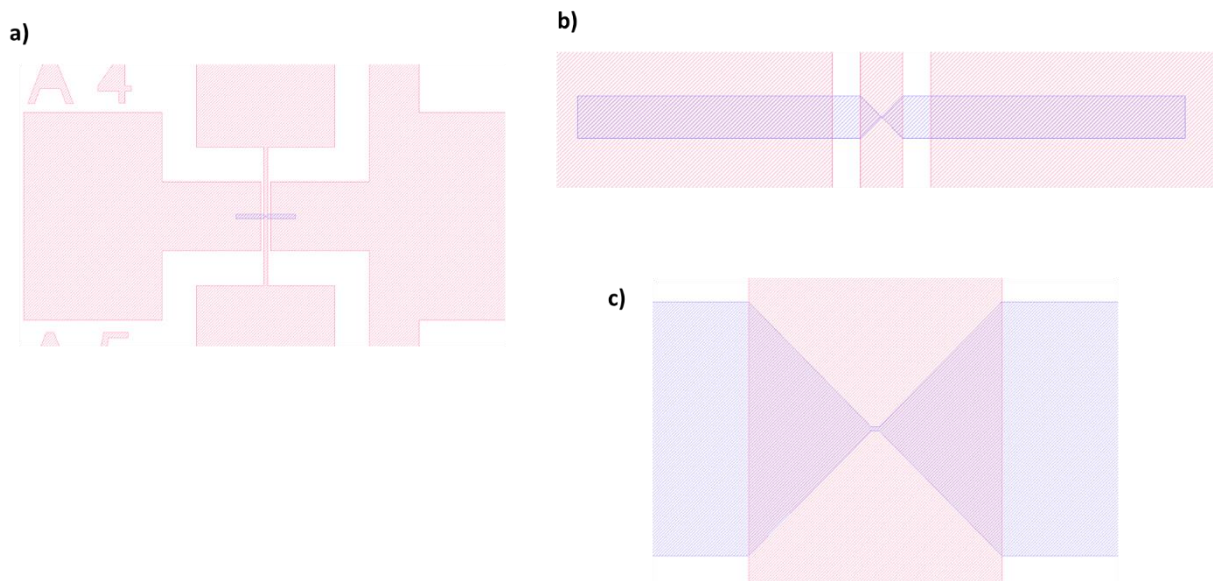


Figure 2 GDS file showing pattern for graphene nanoconstriction on a single device.

Alignment

The narrowest part of the nanoconstriction must be aligned between the source-drain electrodes so that it sits on top of the gate electrode. Alignment accuracy is approximately $0.5\ \mu\text{m}$.

There are multiple sets of marks for use. I generally use the $3\ \mu\text{m}$ width marks in the central groups towards each edge of the chip. Please refer to the GDS file for their coordinates. In the GDS file the alignment marks are in layer 2/0. See figure 3 for more info.

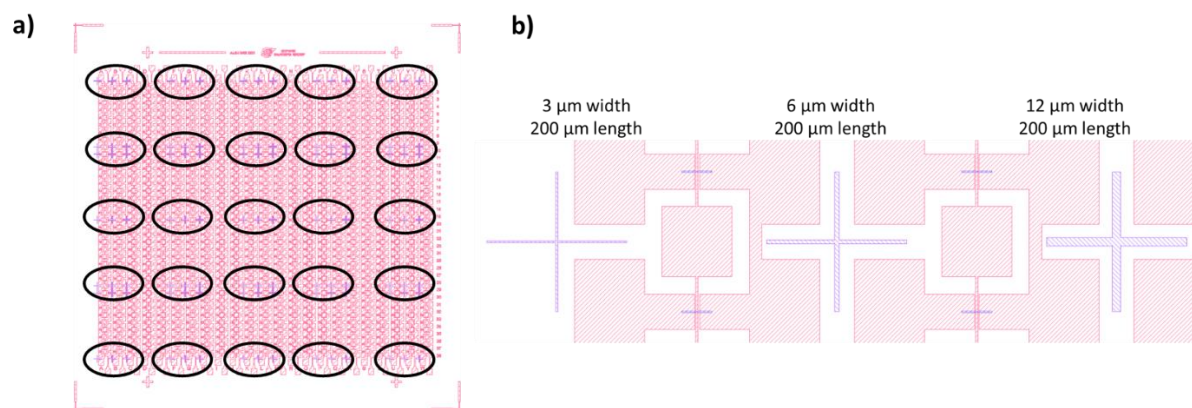


Figure 3 a) Alignment mark location on the chip (black circles). b) alignment mark dimensions.

Fabrication procedure

I suggest we do the following for the fabrication:

1. I will send three blank chips coated with ARN-7500.
2. Dose test, 10×20 array of the nanoconstrictions, separated by $100\ \mu\text{m}$ in X and $200\ \mu\text{m}$ in Y. Do a dose test with a starting dose of $50\ \mu\text{C}/\text{cm}^2$ to final dose of $750\ \mu\text{C}/\text{cm}^2$ (each varied by 3.5

$\mu\text{C}/\text{cm}^2$). Can you place an identifying mark so I can work out what dose corresponds to which feature?

3. Send the blank exposed chips to me for development. I will SEM and the find optimum dose.
4. I will send chips coated in resist for the exposure.
5. You do the exposure and send them back to me for development and further processing.

Let me know if there are any questions or suggestions or if I have missed anything.